Practical No. __

Title: Semantic Segmentation using U-Net on Oxford-IIIT Pet Dataset

Objective

The objective of this experiment is to understand the concept of semantic segmentation and implement a U-Net-based deep learning model for segmenting images from the Oxford-IIIT Pet Dataset. The experiment aims to preprocess images and segmentation masks, train a segmentation network for pixel-wise classification, visualize the predicted segmentation outputs, compare them with the corresponding ground truth masks, and evaluate the performance using standard segmentation metrics.

Problem Statement

The task is to develop a deep learning model capable of performing semantic segmentation on pet images using the Oxford-IIIT Pet Dataset. The model should accurately classify every pixel in an image into its corresponding category and generate meaningful segmentation masks. The generated masks must then be compared with the ground truth masks provided in the dataset, and the segmentation quality should be evaluated using IoU, Dice Coefficient, and Pixel Accuracy metrics.

Course Outcomes Addressed

This practical helps achieve Course Outcome CO1 by providing knowledge of deep learning and semantic segmentation concepts. It supports Course Outcome CO2 through the design and implementation of the U-Net architecture for image segmentation. It also contributes to Course Outcome CO4 by evaluating model performance using standard segmentation evaluation metrics and analyzing the quality of segmentation results.

Software and Hardware Requirements

The implementation of this practical requires Python programming language along with libraries such as TensorFlow, Keras, NumPy, Matplotlib, OpenCV, and TensorFlow Datasets. The development environment may be Jupyter Notebook, Google Colab, or Anaconda. The hardware requirements include a computer with at least an Intel Core i5 processor, 8 GB RAM, sufficient storage space for the dataset and trained model, and preferably an NVIDIA GPU for faster model training.

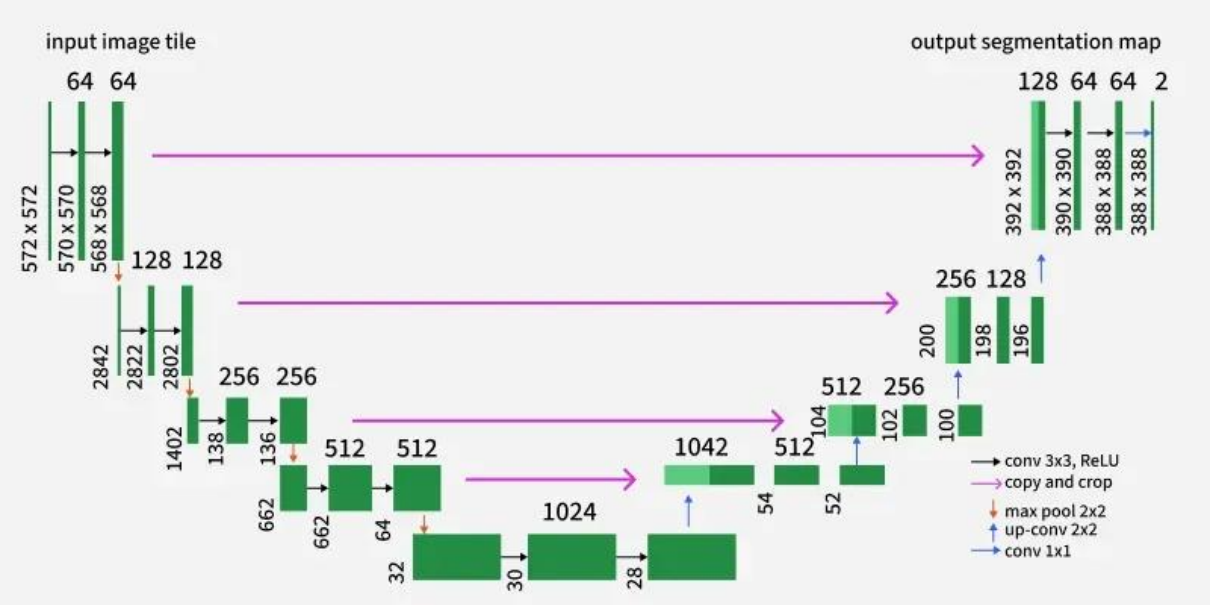
Theory Concept

Semantic segmentation is a computer vision technique in which each pixel of an image is assigned a category label. The primary objective is to obtain a dense prediction map that accurately represents different regions of the image. Unlike object detection, which predicts bounding boxes around objects, semantic segmentation provides a detailed understanding of object shapes and boundaries. This technique is widely used in autonomous driving, medical image analysis, robotics, satellite image processing, and scene understanding applications.

The Oxford-IIIT Pet Dataset is a popular benchmark dataset used for image segmentation and classification tasks. It contains over seven thousand images of cats and dogs spanning thirty-seven different pet breeds. Each image is accompanied by a segmentation mask that identifies foreground objects, background regions, and boundaries. These annotated masks make the dataset suitable for training segmentation models.

U-Net Architecture

U-Net is a fully convolutional neural network architecture specifically designed for image segmentation. The architecture consists of two major components known as the encoder and decoder. The encoder, also called the contracting path, extracts features from input images through convolution and pooling operations. As the network progresses deeper, the spatial dimensions decrease while feature richness increases. The decoder, also known as the expanding path, restores spatial information through upsampling operations. One of the key characteristics of U-Net is the presence of skip connections, which transfer feature maps from the encoder to the decoder. These connections help preserve spatial details that might otherwise be lost during downsampling. As a result, U-Net can produce highly accurate segmentation masks with well-defined object boundaries.



The pixel-wise classification process involves assigning a class label to every pixel in the image. During training, the model learns to map input image pixels to their respective mask labels. The objective function minimizes the difference between predicted masks and actual masks, allowing the network to progressively improve segmentation performance with each training iteration.

Algorithm

The experiment begins by downloading the Oxford-IIIT Pet Dataset and loading both images and their corresponding segmentation masks. Each image and mask are resized to a fixed dimension to ensure uniformity during training. The image pixel values are normalized to

improve convergence and reduce training instability. The segmentation masks are preprocessed to generate valid class labels suitable for training.

After preprocessing, the U-Net architecture is constructed. The encoder portion extracts hierarchical features through stacked convolutional and pooling layers, while the decoder reconstructs high-resolution segmentation maps through upsampling and concatenation with encoder features. The model is compiled using an appropriate optimizer and loss function designed for multiclass pixel classification.

The network is then trained using the prepared training dataset. During training, the model learns to identify pet regions and boundaries by minimizing segmentation errors. Once training is completed, the model predicts segmentation masks for unseen test images. The predicted masks are visualized alongside the original images and corresponding ground truth masks to assess visual quality. Finally, quantitative performance evaluation is carried out using IoU, Dice Coefficient, and Pixel Accuracy metrics.

Explanation of the Program

The program begins by importing the required libraries and loading the Oxford-IIIT Pet Dataset using TensorFlow Datasets. A preprocessing function resizes all images and segmentation masks to a standard size of 128 × 128 pixels. The image pixels are normalized by dividing them by 255 so that values lie between 0 and 1. The segmentation labels are adjusted to suit the classification task.

The U-Net architecture is then defined. The encoder progressively extracts feature representations using convolution and pooling layers. A bottleneck layer captures high-level semantic information. The decoder reconstructs the segmentation mask using upsampling operations and skip connections. The final convolutional layer uses a softmax activation function to predict the probability of each class for every pixel.

The model is compiled using the Adam optimizer and sparse categorical cross-entropy loss function. Training is performed for several epochs, allowing the model to learn segmentation patterns from the dataset. After training, predictions are generated for test images and used for visualization and evaluation.

Visualization of Results

The visualization step displays the original pet image, its corresponding ground truth mask, and the predicted segmentation mask. This comparison helps in understanding how accurately the model identifies object boundaries and foreground regions. A successful segmentation model should generate masks that closely resemble the annotated ground truth masks.

Evaluation Metrics

The performance of a semantic segmentation model cannot be measured solely by classification accuracy because segmentation requires pixel-level correctness. Therefore, specialized evaluation metrics are used.

Intersection over Union (IoU) measures the overlap between the predicted segmented region and the actual segmented region. It is calculated as the ratio of the intersection area to the union area of predicted and ground truth masks. Higher IoU values indicate better segmentation performance.

The Dice Coefficient measures the similarity between predicted and actual segmentation masks. It emphasizes correctly identified pixels and is particularly useful for segmentation tasks involving imbalanced classes. A Dice score closer to one indicates strong agreement between prediction and ground truth.

Pixel Accuracy measures the percentage of pixels correctly classified by the model. It is computed as the ratio of correctly predicted pixels to the total number of pixels in the image. Higher pixel accuracy indicates effective segmentation but should be interpreted alongside IoU and Dice Coefficient because it may be influenced by class imbalance.

Result and Conclusion

In this practical, semantic segmentation was successfully implemented using the U-Net architecture and the Oxford-IIIT Pet Dataset. The dataset images and masks were preprocessed before training the model for pixel-wise classification. The trained model generated segmentation masks that accurately identified pet regions and background regions. Visualization of predictions confirmed that the U-Net architecture effectively captured object boundaries and spatial details. Performance evaluation using Intersection over Union, Dice Coefficient, and Pixel Accuracy demonstrated the effectiveness of the segmentation model. Therefore, it can be concluded that U-Net is a highly suitable deep learning architecture for semantic segmentation tasks and provides reliable segmentation performance on the Oxford-IIIT Pet Dataset.